MIL-J-83010B(USAF)
16 OCTOBER 1987
SUPERSEDING
MIL-J-83010A(USAF)
20 April 1967

MILITARY SPECIFICATION

JACK, AIRCRAFT LANDING GEAR MMU-87/E, 35-TON CAPACITY

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of wheel mounted, 35-ton capacity hydraulic axle jack, designated MMU-87/E.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated Plywood

MILITARY

MIL-P-116	Preservation, Methods of
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance.
MIL-H-81019	Hydraulic Fluid, Petroleum Base, Ultra-low Temperature

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1730

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:32Z
Check the source to verify that this is the current version before use.

MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-147	Palletized Unit Loads
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing, with Appropriate Test Methods
MIL-STD-2073/1A	DOD Materiel Procedures for Development and Application of Packaging

* 2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

AIR FORCE

65J1591	Jack, Hydraulic, Hand, Axle, 35 Ton Capacity, Type MMU-87/E

* (Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issues of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM D-3951	Packaging, Commercial

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:32Z
Check the source to verify that this is the current version before use.

3.2 General requirements. The requirements of drawing 65J1591 apply as requirements of this specification. When the drawing and this specification conflict, this specification shall govern.

3.3 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.4 Workmanship. The jack, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges, accuracy of dimensions, and marking of parts and assemblies.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.3 Test conditions. The jack shall be serviced with hydraulic fluid conforming to MIL-H-5606 except for the low temperature test which shall be conducted with fluid conforming to MIL-H-81019.

4.3.1 Hydraulic fluid pumping source. The integral hydraulic pumping system shall be operated for load lifting tests specified herein.

4.4 First article inspection (see 6.2).

Source: https://assist.dla.mil Downloaded: 2026-10-23T07:32Z
Check the source to verify that this is the current version before use.

4.4.1 Test sample. One jack shall be subjected to the tests specified in 4.4.3.

4.4.2 Test report. After completion of the first article test, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished the procuring activity.

4.4.3 First article tests. The first article tests shall consist of all tests described under 4.6.

4.5 Quality conformance inspection. The acceptance tests shall consist of the following:

a. Individual tests.

b. Sampling plan and tests.

4.5.1 Individual tests. Each jack shall be subjected to the following tests as described under 4.6:

a. Examination of product.

b. Ram and cylinder hydrostatic pressure.

c. Pump bypass relief valve test.

4.5.2 Sampling plan and tests.

4.5.2.1 Periodic sampling. Five jacks shall be selected at random from every fifty or fraction thereof produced and subjected to the static proof test as described under 4.6.

4.5.2.1.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items on hand or produced later shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all tests shall be repeated.

4.5.2.1.2 Individual tests may continue. For production reasons, individual test may be continued pending the investigation of a test failure. But final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of this specification.

4.5.3 Defects in items already accepted. The investigation of the test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the methods of correcting them.

4.6 Test methods.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:32Z
Check the source to verify that this is the current version before use.

4.6.1 Examination of product. The jack shall be inspected to determine compliance with the requirements specified herein with respect to workmanship, marking, and compliance with the drawing.

4.6.2 Ram and cylinder hydrostatic pressure. The ram and cylinder assembly shall be proof-tested at 7,500 psi hydraulic pressure. This test shall be performed with an appropriate load or restraint on the ram assembly to preclude damage to the ram stop. Any leakage of hydraulic fluid shall be cause for rejection.

4.6.2.1 Settling. With the jack in the fully extended position and supporting the rated capacity load, the jack shall be allowed to stand for 1 hour. The height shall be measured at the beginning and at the end of this period. The rate of settling of the jack shall not exceed 0.020 inch per hour under rated capacity load, any excess of this distance shall be cause for rejection.

4.6.3 Pump bypass relief valve. The pump shall be proof-tested by sealing the pump outlet and operating the pump until the bypass valve opens. The bypass valve shall be adjusted as specified on the drawing.

4.6.4 Life test. With the screw extension retracted, the jack shall be subjected to the following tests:

4.6.4.1 Vertical lift. The vertical lift test shall consist of 200 cycles of raising and lowering a vertical load of 35 tons. One cycle shall consist of raising the hydraulic ram under load from the fully retracted position to the fully extended position and return to the fully retracted position. Failure of any part of the jack during this test shall be cause for rejection.

4.6.4.2 Dynamic side load. The dynamic side load test shall consist of 50 cycles of raising a 35-ton load vertically with a graduated 5.25 horizontal load. The horizontal load shall be applied through the jack pad socket adapter along the centerline of the major and minor axes with respect to the jack base. Twenty-five cycles shall be directed to each axis. The horizontal load on the jack adapter shall be gradually increased from zero in the fully retracted position to 5.25 tons in the fully extended position. Failure of any part of the jack during this test shall be cause for rejection.

4.6.5 Static proof. After the bypass relief valve has adjusted for overload, the jack shall be tested with the screw extension in the fully extended position by raising a 52.5 ton load vertically from the lowest position of the hydraulic ram to the fully extended position. While in the extended position, the vertical load shall be reduced to 35 tons and maintained during application of a 5.25 ton horizontal load applied to the top of the extended screw extension. To complete the cycle, the horizontal and vertical loads shall be removed and the jack ram allowed to retract to its original position with no more than 50 pounds applied to the top of the ram. Failure of any part of the jack during this test shall be cause for rejection.

4.6.6 Low temperature. The jack shall be subjected to low temperature in accordance with MIL-STD-810, method 502, procedure I. A temperature of -54°C shall be maintained for a period of 24 hours. At the end of the exposure period and with the temperature maintained, the jack ram shall be raised to its maximum extension and pumped until the bypass relief valve operates. The lowering valve shall be opened, and the ram shall retract with not more than 50 pounds applied to the ram. For this test, the jack shall be serviced with

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:32Z
Check the source to verify that this is the current version before use.

hydraulic fluid conforming to MIL-H-81019. The time and number of full strokes necessary to fully extend the ram shall be recorded at both +21° and -54°C. The time and strokes required to fully extend the ram at -54°C shall be not more than double the time and strokes required at +21°C.

4.6.7 High temperature. The jack shall be subjected to high temperature in accordance with MIL-STD-810, method 501, procedure I. At the end of the exposure and with the temperature maintained, the jack ram shall be raised to its maximum extension and pumped until the bypass relief valve operates. The lowering valve shall be opened and ram shall retract with not more than a 50-pound force applied to ram. During this test, there shall be no evidence of external leakage of hydraulic fluid from the ram-cylinder assembly.

4.7 Inspection of packaging. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PACKAGING

* 5.1 Preservation. Preservation shall be level A, C or commercial as specified (see 6.2).

* 5.1.1 Level A. Unless otherwise specified in the contract or order, each jack shall be individually preserved in accordance with method I of MIL-P-116.

* 5.1.1.1 Preservation application. Preservatives in accordance with MIL-P-116 shall be used.

* 5.1.2 Level C. The level C preservation for the hydraulic jack shall conform to the MIL-STD-2073 requirements for this level.

* 5.1.3 Commercial. The commercial preservation of the hydraulic jack shall be in accordance with the requirements of ASTM D-3951.

* 5.2 Packing. Packing shall be level A, B, C or commercial as specified (see 6.2).

* 5.2.1 Level A. Each jack preserved as specified in 5.1 shall be packed in a container conforming to PPP-B-601, style A or B, overseas type, with treated lumber and plywood.

* 5.2.2 Level B. Each jack preserved as specified in 5.1 shall be packed in a container conforming to PPP-B-601, style A or B, overseas.

* 5.2.3 Level C. The level C packing for the jack shall conform to the MIL-STD-2073 requirements for this level.

* 5.2.4 Commercial. The preserved jack shall be packed in accordance with the requirements of ASTM D-3951.

* 5.2.5 Palletized loads. When specified (see 6.2) palletized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:32Z
Check the source to verify that this is the current version before use.

* 5.3 Marking.

* 5.3.1 Levels A, B and C. In addition to any special or other identification marking required by the contract (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

* 5.3.2 Commercial. Commercial markings shall be in accordance with the requirements of ASTM D-3951.

* 5.4 General.

* 5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, 5.2.3 and 5.2.4) shall be of minimum tare weight and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

* 5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.1.

* 5.4.3 Physical protection. Interior blocking, bracing, cushioning, et cetera, shall be in accordance with MIL-STD-1186.

6. NOTES

6.1 Intended use. The MMU-87/E jack is intended for use as an axle jack for aircraft whose jack pads are between 9 1/2 and 22 1/2 inches from the ground.

* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Location and conditions for first article testing (see 4.4).

c. Applicable level of preservation, packaging, and packing (see 5.1 and 5.2).

* d. When palletized loads are required (see 5.2.5).

6.3 First article. When a first article inspection is required, the item(s) should be a first article sample. The first article should consist of one, unit(s). The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:32Z
Check the source to verify that this is the current version before use.

6.4 Subject term (key word) listing.

Aircraft landing gear
Axle
Capacity, 35-ton
Hand operated
Hydraulic jack
Hydrostatic pressure
Jack
MMU-87/E
Ram and cylinder
Wheel mounted

6.5 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

Project 1730-F291

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:32Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 73236 WASH DC

POSTAGE WILL BE PAID BY HQ U.S. AIR FORCE

San Antonio ALC/MMEDO
Kelly AFB, Tx 78241

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-J-83010	Jack Aircraft Landing Gear, MMU-87E, 35-Ton Capacity

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:32Z
Check the source to verify that this is the current version before use.